

MENDSPEECH REVISED LEARNING SYSTEM

## **Week 6: Robustness, Fine Tuning, RNNT, and Calibration**

Adapt the recognizer to damaged speech while learning training and calibration discipline.

Six focused build days plus one consolidation day. Each day ends with named artifacts or measured evidence.

## Week milestone

Adapt the recognizer to damaged speech while learning training and calibration discipline.

## Week map

| Day    | Focus                                       | Minimum evidence                                                                               | Compute                                                              |
|--------|---------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Day 36 | Training pipeline anatomy                   | • Deliberately use a bad learning rate and record the failure signature.                       | Modal L4                                                             |
| Day 37 | Build a robust fine tuning dataset          | • Audit duplicate and speaker leakage.                                                         | Local CPU                                                            |
| Day 38 | Fine tune for damaged speech robustness     | • Compare base and adapted model on the frozen test set.                                       | Modal L4, consider L40S only if memory blocks the planned experiment |
| Day 39 | SpecAugment and augmentation ablation       | • Compare no augmentation versus selected augmentation with the same seed and training budget. | Modal L4                                                             |
| Day 40 | RNNT and streaming decoding                 | • Compare CTC and transducer outputs on selected difficult clips.                              | Modal L4                                                             |
| Day 41 | Confidence calibration for repair decisions | • Compare raw and calibrated confidence if a simple method is feasible.                        | Modal L4 for logits, local CPU for analysis                          |
| Day 42 | Week 6 robustness milestone                 | • Run one fixed benchmark suite and freeze results for Week 8 comparisons.                     | Modal L4                                                             |

## Reference spine

- NVIDIA NeMo ASR training documentation
- RNNT primary references
- Calibration and reliability diagram references

# Training pipeline anatomy

## LEARN

- Manifest format.
- Batching variable duration audio.
- Loss curves.
- Learning rate.
- Validation split.
- Checkpointing.

## BUILD IN MENDSPEECH

- Create a tiny reproducible training configuration.
- Run a short smoke training job and verify loss decreases.

## EXPERIMENT AND MEASURE

- Deliberately use a bad learning rate and record the failure signature.

## REQUIRED OUTPUT

- configs/train\_smoke.yaml
- results/day36\_training\_smoke.csv
- docs/training\_debug\_notes.md

**Completion check:** You can diagnose whether a run is learning, diverging, or overfitting from basic evidence.

### Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

## REFERENCE PRIORITY

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# Build a robust fine tuning dataset

## LEARN

- Train, validation, test separation.
- Speaker leakage.
- Synthetic corruption sampling.
- Balanced severity distribution.

## BUILD IN MENDSPEECH

- Create manifests that pair clean transcripts with corrupted audio.
- Keep a speaker separated test set frozen.

## EXPERIMENT AND MEASURE

- Audit duplicate and speaker leakage.

## REQUIRED OUTPUT

- data/train\_manifest.jsonl
- data/val\_manifest.jsonl
- data/test\_manifest.jsonl
- reports/data\_audit.md

**Completion check:** The evaluation set cannot accidentally appear in training through clean or corrupted duplicates.

### Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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# Fine tune for damaged speech robustness

**Compute**

Modal L4, consider L40S only if memory blocks the planned experiment

**LEARN**

- Transfer learning.
- Frozen versus trainable layers.
- Mixed precision.
- Gradient accumulation.

**BUILD IN MENDSPEECH**

- Fine tune a manageable FastConformer or compatible ASR checkpoint on the robustness dataset.
- Save model, config, and training logs.

**EXPERIMENT AND MEASURE**

- Compare base and adapted model on the frozen test set.

**REQUIRED OUTPUT**

- training/finetune.py
- checkpoints/week6\_best/
- results/day38\_base\_vs\_adapted.csv

**Completion check:** You can state exactly what improved, what did not, and whether clean speech regressed.

**Study method**

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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# SpecAugment and augmentation ablation

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## LEARN

- Time masking.
- Frequency masking.
- Data augmentation as invariance training.

## BUILD IN MENDESPEECH

- Add one augmentation intervention to a controlled short run.

## EXPERIMENT AND MEASURE

- Compare no augmentation versus selected augmentation with the same seed and training budget.

## REQUIRED OUTPUT

- experiments/specaugment\_ablation.py
- results/day39\_augmentation.csv

**Completion check:** You can separate the effect of augmentation from the effect of extra training time.

### Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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# RNNT and streaming decoding

## LEARN

- Encoder.
- Prediction network.
- Joint network.
- Blank handling.
- Streaming emission behavior.
- Difference from CTC independence.

## BUILD IN MENDSPEECH

- Run or inspect a FastConformer transducer checkpoint.
- Trace one decoding step conceptually and document tensor roles.

## EXPERIMENT AND MEASURE

- Compare CTC and transducer outputs on selected difficult clips.

## REQUIRED OUTPUT

- docs/rnnt\_walkthrough.md
- results/day40\_ctc\_vs\_rnnt.csv

**Completion check:** You can explain RNNT without saying only that it is better for streaming.

### Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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# Confidence calibration for repair decisions

**Compute**

Modal L4 for logits, local CPU for analysis

**LEARN**

- Reliability diagrams.
- Expected calibration error intuition.
- Threshold selection from validation data.

**BUILD IN MENDSPEECH**

- Build a simple calibration analysis for confidence versus correctness.
- Choose policy thresholds on validation, not test.

**EXPERIMENT AND MEASURE**

- Compare raw and calibrated confidence if a simple method is feasible.

**REQUIRED OUTPUT**

- src/asr/calibration.py
- results/day41\_reliability.png
- configs/repair\_modes\_calibrated.yaml

**Completion check:** Repair thresholds are now justified from held out evidence rather than guessed.

**Study method**

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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# Week 6 robustness milestone

## LEARN

- Review fine tuning, augmentation, RNNT, and calibration.

## BUILD IN MENDSPEECH

- Switch between base and adapted recognizer in the research console.
- Show clean WER, damaged WER, confidence calibration, and repair percentage.

## EXPERIMENT AND MEASURE

- Run one fixed benchmark suite and freeze results for Week 8 comparisons.

## REQUIRED OUTPUT

- app/mendspeech\_v3\_robust.py
- results/week6\_frozen\_baseline.csv
- reports/week6\_training.md

**Completion check:** MendSpeech can demonstrate measured robustness gains or clearly document a negative result.

### Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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